Summary Sheet: Recirculating Water-Cooling Tower

Volume (kg/yr):

1. Concentration of PMN in recirculating water: $[x_R]$ = ppm
   Submitter estimate: ______ ppm
   Default values scale control: 1 to 5 ppm
   corrosion control: 50 to 1000 ppm
   Other information: ppm (Source:)

2. Recirculation rate of cooling water: [R] = gal/minute (gpm)
   Submitter estimate: __________ gpm
   Default value Moderately-sized Tower: 2,000 gpm
   Large tower: 100,000 gpm
   Other information: __________ gpm (Source:)

3. Water Releases from Blowdown: [B] = kg/site/day
   Assumptions
   360 days/yr of release
   discharge to onsite wwt or POTW
   Water release equal to 0.6% of the recirculation rate
   [Note: typical range is 0.5 to 0.6%].

   $B = 0.6\% \times x_R \times R \times (5760 \times 10^{-6}\ \text{min-}kg/hr\text{-}gal)$

   $B = 3.5 \times 10^{-5} \times x_R \times R$

   $B =$ __________ $kg/site/day$

4. Air Releases from Windage or Entrainment: [W] = kg/site/day
   Assumptions
   Evaporative losses are negligible/ PMN is nonvolatile.
   360 days/yr of release
   Windage losses release equal to 0.1% of the recirculation rate [Note: typical range is 0.008 to 0.2%].

   $W = 0.1\% \times x_R \times R \times (5760 \times 10^{-6}\ \text{min-}kg/hr\text{-}gal)$

   $W = 6 \times 10^{-6} \times x_R \times R$

   $W =$ __________ $kg/site/day$

5. Estimate of Throughput: [T]= kg/site/year

   $T = (B+W) \times 360 =$ __________ $kg/site/day$

6. Estimate/Check Number of Sites

   $$\text{number of sites} = \frac{Volume}{T} =$$

Recirculating Water-Cooling Tower

PMN chemicals are frequently used as water treatment additives in recirculating cooling systems to prevent corrosion, scaling, and growth of microorganisms. Typical concentrations of additives in the recirculating water are listed in Table 1.

Table 1: Cooling Water Additives[1]

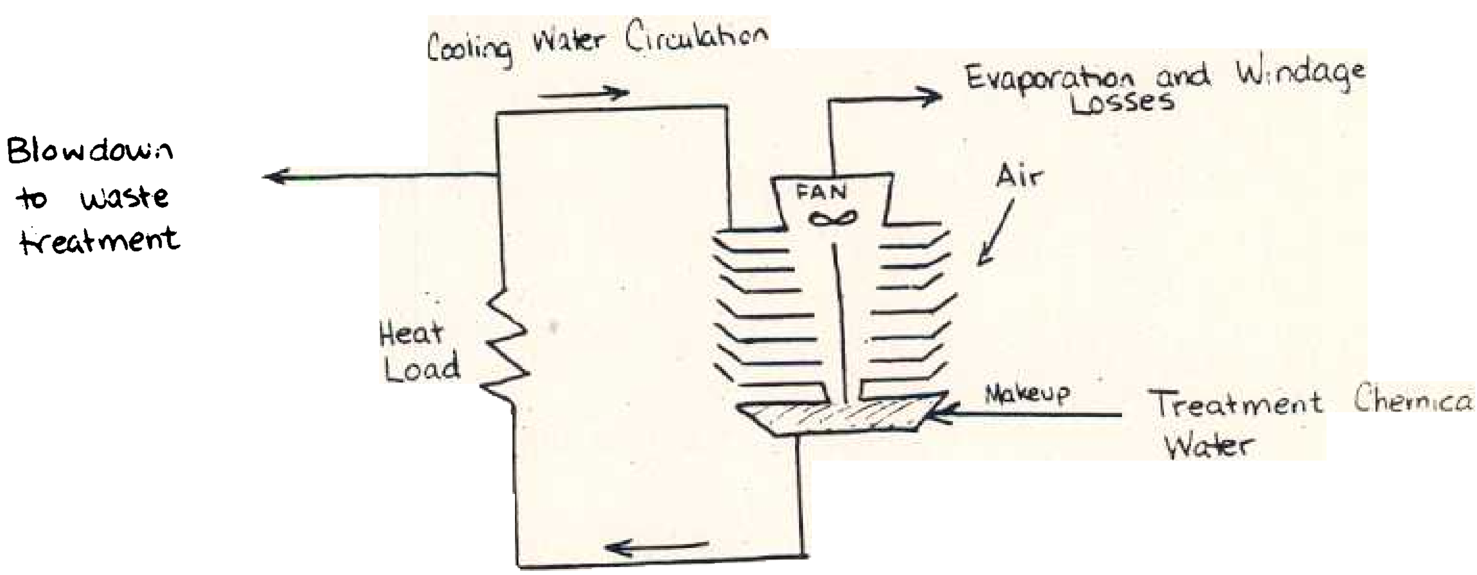
Type of Additive	Typical Concentration (ppm)
Scale Control	1 to 5
Corrosion Control	50 to 1000
Microorganism Control[2]	1 to 20

[1] Source: Keifer LC, et al, no date, pp 24 to 28.
[2] This is a FIFRA use and not generally covered by TSCA.

CEB typically assumes a recirculation rate of 2,000 gallons per minute (gpm) for a moderately-sized tower. A large cooling system may have a recirculation rate of up to 100,000 gpm.

Makeup fresh water must be added to the system to replace losses from evaporation, entrainment (drift or windage), and blowdown. Releases from a recirculating cooling tower are shown in Figure 1.

Figure 1: Schematic of Cooling Tower System



Description of Releases

Evaporative losses typically range from 0.5 to 3% of the recirculation rate. If the PMN chemical is non-volatile, the losses of PMN chemical due to evaporation are expected to be negligible.

Evaporation of water causes the nonvolatile materials to accumulate. To keep the salt concentration at a predetermined level, a small amount of water is deliberately discarded (blowdown). CEB generally assumes that blowdown is about 0.5 to 0.6% of the recirculation rate. Typically, blowdown is sent to either an on-site treatment plant or POTW.

Windage losses are a function of the mist eliminator design and generally range from less than 0.1% to up to 0.2% of the recirculation rate. Some cooling tower manufacturers warrant as low as 0.008% for windage losses. In the absence of other information, CEB assumes windage losses to be 0.1% of the recirculation rate.

Calculations

Assuming that the blowdown is equal to 0.6% of the recirculation rate, and converting to the appropriate units, the water releases are estimated as :

$$B = 0.6\% \times x_r \times R \times (5760\times10^{-6}\ \text{min-kg/hr-gal})$$

where B = *blowdown (kg/site/day)*
x_r = *concentration of PMN chemical (ppm)*
R = *recirculation rate (gpm)*

Assuming that the recirculation rate is 2000 gpm and that releases will occur over 360 days/yr, releases to onsite wwt or POTW are estimated as:

$$B = 0.07 \times x_r$$

where B = *blowdown (kg/site/day)*
x_r = *concentration of PMN chemical (ppm)*
R = *recirculation rate (gpm)*

Assuming that windage is equal to 0.1% of the recirculation rate and converting to the appropriate units, air releases are estimated as:

$$W = 0.1\% \times x_r \times R \times (5760 \times 10^{-6}\ \text{min-}kg/hr\text{-}gal)$$

where W = windage (kg/site/day)
x_r = concentration of PMN chemical (ppm)
R = recirculation rate (gpm)

Assuming that the recirculation rate is 2000 gpm and that releases will occur over 360 days/yr, releases to onsite wwt or POTW are estimated as:

$$W = 0.012 \times x_r$$

where W = windage (kg/site/day)
x_r = concentration of PMN chemical (ppm)
R = recirculation rate (gpm)

Makeup or throughput of the PMN chemical is estimated as the total of the windage and blowdown losses.

References

Keifer LC, et al. n.d. Industrial Process Profiles to Support PMN Review: Waste Treatment Chemicals. Walk,Haydel&Associates. EPA COntract 68-01-6065.

Kunz RG, et al. "Cooling-water Calulations" Chemical Engineering vol. 84, no 16, August 1, 1977 p61-71.

Perry, RH D Green Editor. Perry's Engineer's Handbook 6th Edition McGraw-Hill, NY, 1984.